SMART POTHOLE DETECTION SYSTEM USING GPS
AND VIBRATION SENSORS

Presented by: Logic Squad

Slide 1: Title

Smart Pothole Detection System Using GPS and Vibration Sensors on Government Vehicles

Group Name: Logic Squad

Project Theme: Smart Cities | E-Government | Road Infrastructure

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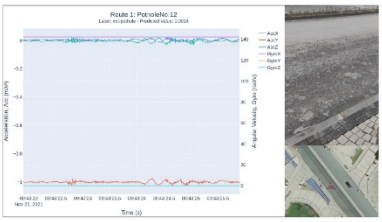
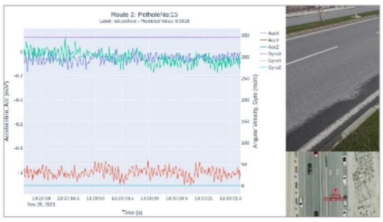
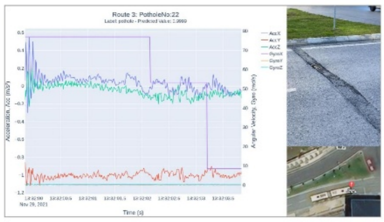
Slide 2: Introduction

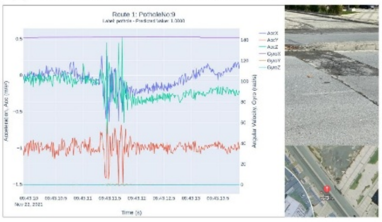
Road potholes are a major problem in many cities. They cause accidents, damage vehicles, increase traffic congestion, and cost governments millions in repairs every year.

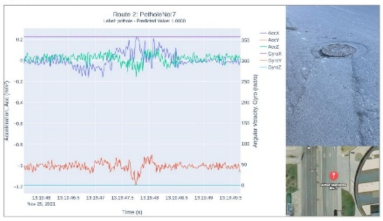
Our project proposes a smart system that automatically detects potholes using sensors installed on government vehicles. The detected potholes are sent to an e-government platform, allowing municipalities to repair roads faster and more efficiently.

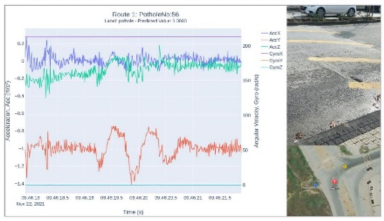

(a) State O: Not Pothole, Route 1, PNo: 12


(b) State O: Not Pothole, Route 2, PNo: 15


(c) State A: Pothole, Route 3, PNo: 22


(d) State A: Pothole, Route 1, PNo: 9


(e) State A: Pothole, Route 2, PNo: 7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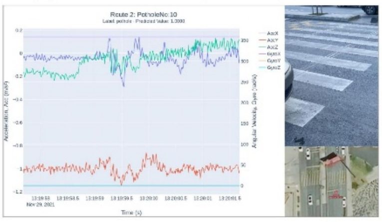
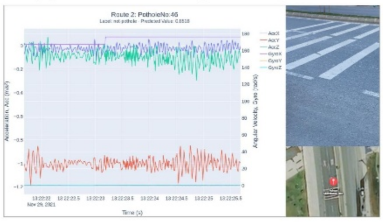

(g) State B: Not Pothole, Route 2, PNo: 10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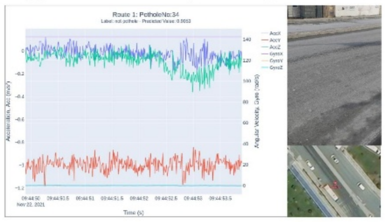

(i) State E: Not Pothole, Route 1, PNo: 34

Slide 3: Problem Statement

Current pothole detection methods have several challenges:

- Citizens report potholes late or not at all.
- Manual road inspections take a long time.
- Repairs are often delayed.
- Some potholes become dangerous before they are reported.
- Government lacks real-time information on road conditions.

Slide 4: Proposed Solution

We propose installing:

- GPS Module
- Vibration (Accelerometer) Sensor
- ESP32/Arduino
- GSM or Wi-Fi Module

on government vehicles such as:

- Road maintenance vehicles
- Refuse trucks
- Public buses
- Municipal inspection vehicles

The system automatically detects potholes while vehicles perform their normal duties.

Slide 5: Components Used

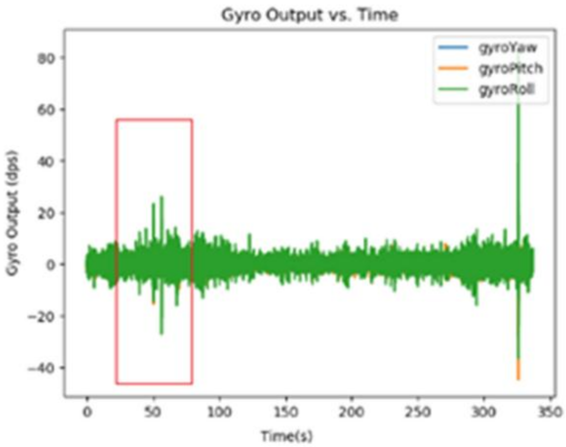
Hardware:

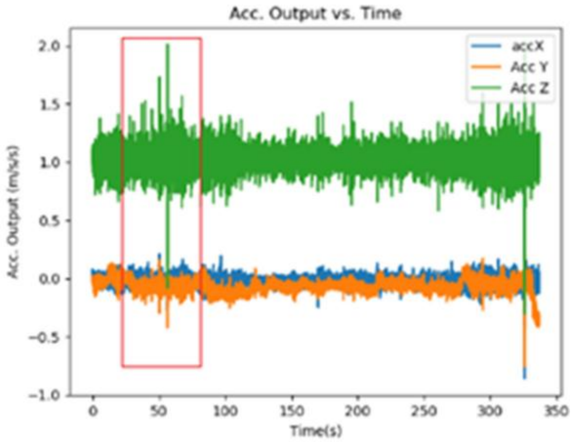
- ESP32 or Arduino Uno
- GPS Module
- Vibration (Accelerometer) Sensor
- GSM/Wi-Fi Module
- Power Supply Software:

- Arduino IDE
- Cloud Database
- E-Government Dashboard

Slide 6: How the System Works

1. Government vehicle drives normally.
2. Vibration sensor detects a sudden impact caused by a pothole.
3. GPS records the exact location.





4. ESP32 processes the data.
5. Data is uploaded automatically.
6. Municipality receives an alert.
7. Repair team is assigned.
8. Public dashboard updates the repair status.

Slide 7: System Architecture

Government Vehicle

Vibration Sensor detects pothole
GPS records location

ESP32 processes data

Internet (Wi-Fi/GSM)

Cloud Database

Municipality Dashboard

Road Maintenance Team

Pothole Repaired

Slide 8: Why Current Methods Are Not Enough

Current Method:

- Manual inspections
- Public complaints
- Delayed reporting
- Limited road coverage
- Human error Our Smart System:

- Automatic detection
- Real-time GPS location
- Faster reporting
- Continuous monitoring
- Accurate records

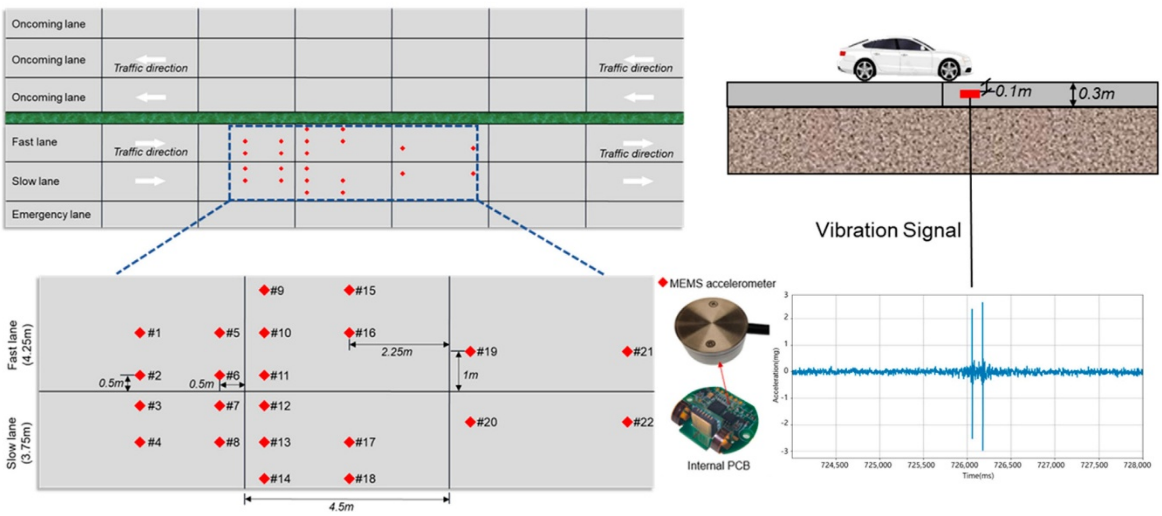
Slide 9: Advantages of GPS and Vibration Sensors

GPS:

- Records exact pothole location.
- Eliminates guesswork.
- Creates digital road maps.

Vibration Sensor:

- Detects bumps immediately.
- Low-cost solution.
- Easy to install.
- Reliable for government fleet vehicles.



Slide 10: Benefits to Government

- Faster pothole detection.
- Lower maintenance costs.
- Better planning.
- Digital records of road conditions.
- Improved service delivery.
- Supports Smart City initiatives.

Slide 11: Benefits to Society

- Safer roads.
- Fewer accidents.
- Less vehicle damage.

- Reduced travel delays.
- Better public transport.
- Increased public trust through transparent repair tracking.

Slide 12: Economic Benefits

Our system contributes to the economy by:

- Reducing vehicle repair costs.
- Lowering emergency road repair expenses.
- Improving transport efficiency.
- Supporting businesses through better roads.
- Increasing productivity.
- Creating technology and maintenance jobs.

Slide 13: Transparency Through E-Government

The dashboard shows:

- Pothole location.
- Date detected.
- Priority level.
- Repair team assigned.
- Repair progress.
- Date completed.

Citizens can see the status of repairs, improving accountability and transparency.

Slide 14: Expected Impact

Detection Speed:

- From days or weeks to minutes or hours.

Accuracy:

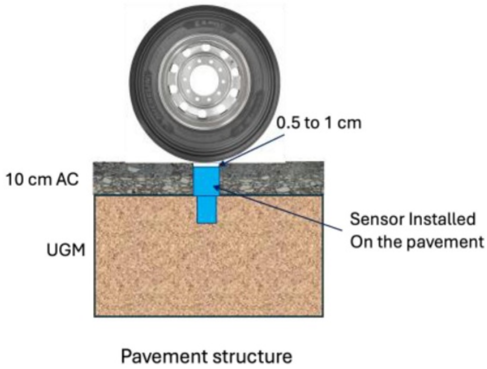
- GPS provides precise locations.
- Sensor data reduces false reports.

Transparency:

- Every pothole is tracked from detection to repair.

Efficiency:

- Government vehicles inspect roads automatically during daily operations.



(a)

(b)

Slide 15: Future Improvements

Future versions may include:

- AI to classify pothole severity.
- Cameras for image verification.
- LiDAR for depth measurement.
- Mobile app for citizens.
- Predictive maintenance using historical data.

Slide 16: Conclusion

Our Smart Pothole Detection System transforms government vehicles into mobile road inspectors.
By combining GPS, vibration sensors, and an e-government dashboard, municipalities can detect potholes earlier, improve road maintenance, reduce costs, enhance road safety, and provide better services to citizens.

This solution supports smarter cities and contributes to economic development through safer and more reliable road infrastructure.

Slide 17: Thank You

Thank You

Questions?

Presented by: **Logic Squad**